

William Chen

Data Platform Engineer

Jersey City, NJ 07302 | wchen1396@gmail.com | willchennn.com | github.com/smallchungus

U.S. Citizen - No Visa Sponsorship Required

EDUCATION

Georgia Institute of Technology

Master of Science in Analytics (Data Science), GPA: 4.00

Atlanta, GA

January 2026 – May 2027

Rutgers University

Bachelor of Arts in Computer Science and Psychology, GPA: 3.44

New Brunswick, NJ

May 2021

EXPERIENCE

Rutgers University

Research Software Engineer

New Brunswick, NJ

August 2022 – Present

- **EURI AI Assistant:** natural-language-to-SQL and **RAG** chatbot over **AWS Athena**, adopted by **20 teams (500-1,000** internal users), retrieving EiMP documentation across datasets with **1,000,000,000+** PII-bearing healthcare rows
- **Production API:** **5 FastAPI** endpoints (health checks, SQL generation, cost estimation, reviewed execution, query statistics), deployed on **AWS ECS Fargate** behind an **ALB**
- **Testing and Reliability:** **127 unit tests** + **84 nonproduction integration tests** (auth, validation, SQL injection, prompt injection, PII enforcement, cost estimation, Athena execution)
- **Security Guardrails:** bearer-token auth, team-based **PII** tiers, **sqlglot** validation, read-only SQL enforcement, prompt sanitization, pre-execution cost checks
- **EiMP Data Quality:** identity quality across **125+** source systems (SSN, DOB, names, phones, emails, MBI, MRN/OID, overmerge patterns) over **1,000,000,000+** rows
- **Identity Linkage:** resolved EID and HealthSafe ID relationships for **23,000,000** ISEC member records via **S3** and **Athena** joins
- **High-Volume Validation:** analyzed a **127,000,000**-record Death Master File feed, validating **~97,800,000** SSN-only and **2,600,000** SSN-and-DOB matches
- **Distributed Identity Messaging:** contributed to SCEPTER and EiMP **Kafka** flows from **125+** source systems through compacted topics with **100+** partitions

Viatric

Data Platform Engineer (Federal Contract)

Arlington, VA

April 2024 – March 2026

- **Scaled USDA analytics to 5,000-10,000 users** by building **50+ Python** and **AWS Glue** ETL pipelines and operating **150+** production pipelines feeding finance, program, and executive dashboards
- **Enabled congressional budget submissions and decisions involving \$100,000,000+** by delivering governed datasets to USDA billing, budget, finance, and program teams
- **Improved visibility into \$1,000,000+ annual cloud spend** by powering dashboards that attributed usage, exposed cost drivers, and guided department-level budgeting
- **Cut pipeline runtime 60% and deployment time from 60 to 10 minutes** using **AWS Lambda**, **Step Functions**, partitioned **Glue** loads, **Terraform**, and **GitHub Actions**
- **Reduced new-feed onboarding time 50%** via reusable pipeline templates, validation patterns, and deployment standards across the **150+** pipeline platform
- **Protected production dashboards** with schema-drift and record-level validation that caught **3** upstream changes in **6 months**
- **Improved reliability across 150+ pipelines and 3 environments** with **Datadog** SLA monitoring, anomaly alerts, **IAM** controls, and on-call runbooks
- **Enabled self-service operations for 100+ internal users** through audit-ready runbooks and platform documentation

TECHNICAL SKILLS

Languages & Databases: Python, SQL, JavaScript, TypeScript, Redis, Amazon Redshift

Cloud & Infrastructure: AWS Glue, Lambda, Step Functions, S3, Athena, ECS Fargate, IAM, Terraform, FastAPI, Kafka, GitHub Actions, Datadog